KCSE Chemistry Marking Scheme & Solution Guide

Subject: Chemistry Paper 1 (80 Marks)
Document Type: Standard Marking Scheme and Model Answers

Question 1: Bunsen Burner & Flames

(a) State how the luminous flame is produced. *(1 mark)*

- It is produced by closing (or partially closing) the air hole of the Bunsen burner so that incomplete combustion occurs. **(1 mark)**

(b) Give one advantage and one disadvantage of a luminous flame compared to the non-luminous flame:

- **(i) Advantage:** It produces light/can be used for lighting purposes. **(1 mark)**
- **(ii) Disadvantage:** It produces a lot of soot (dirty/blackens apparatus) AND/OR produces less heat (is not hot enough). **(1 mark)**

Question 2: Basic Concepts of Matter

(a) State what is meant by the term:

- **(i) Molecule:** The smallest neutral particle of an element or compound that can exist independently and retain chemical properties. **(1 mark)**
- **(ii) Compound:** A substance containing two or more different elements chemically combined together in a fixed ratio. **(1 mark)**

(b) Identify one other characteristic of a mixture. *(1 mark)*

- The constituents retain their individual chemical properties. **OR**
- It has variable composition (constituents can be combined in any proportion by mass). **OR**
- No energy change (heat absorption/evolution) occurs during its formation. **OR**
- It has variable melting/boiling points (sharp melting point is absent). **(1 mark)**

Question 3: Periodic Properties of Period 3 Elements

Table 1 shows properties of elements W, X, Y, and Z belonging to the same period.

(a) Arrange the elements in the order that they appear in the period. *(1 mark)*

- **Order: X → Z → Y → W** *(or X, Z, Y, W)* **(1 mark)**
  (Reason: Across a period from left to right, atomic radius decreases as nuclear charge increases: X (0.186 nm), Z (0.143 nm), Y (0.117 nm), W (0.000 / gas)).

(b) Fill in the table the type of structure present in each element. *(2 marks)*

Element	Atomic Radius (nm)	Melting Point (°C)	Electrical Conductivity	Type of Structure
W	0.000 (gas)	-101	Poor	**Simple molecular structure** *(½ mark)*
X	0.186	98	Conducts	**Giant metallic structure** *(½ mark)*
Y	0.117	1410	Poor	**Giant atomic / Giant covalent structure** *(½ mark)*
Z	0.143	660	Conducts	**Giant metallic structure** *(½ mark)*

Question 4: Acids, Bases and Neutralization

(a) When a student was stung by a bee, the lab technician applied aqueous ammonia to the affected area and the student was relieved of pain. Explain the process. *(2 marks)*

- Bee venom is acidic (contains methanoic acid). **(1 mark)**
- Aqueous ammonia is a weak base/alkaline solution which neutralizes the acid, thereby relieving pain. **(1 mark)**

(b) Name another solution that the technician could have used. *(1 mark)*

- Sodium hydrogen carbonate solution (Sodium bicarbonate / Baking soda solution) **OR** Calcium hydroxide (Limewater) / Magnesium hydroxide (Milk of Magnesia). **(1 mark)**

Question 5: Stoichiometry & Organic Formula Determination

Empirical formula = CH_2Br. Mass = 45.1 g occupying 5.76 dm^3 at r.t.p. (Molar gas volume = 24.0 dm^3).

(a) Determine the molecular formula of compound V. *(2 marks)*

Step 1: Calculate moles of V:

$$Moles = \frac{Volume}{MolarGasVolume} = \frac{5.76dm^3}{24.0dm^3/mol} = 0.24moles$$

- *(½ mark)*

Step 2: Calculate Molar Mass of V:

$$MolarMass = \frac{Mass}{Moles} = \frac{45.1g}{0.24moles} \approx 187.92g/mol \approx 188g/mol$$

- *(½ mark)*

Step 3: Calculate Empirical Formula Mass (EFM):

$$EFMofCH_2Br = 12.0 + (2 \times 1.0) + 80.0 = 94.0g/mol$$

- *(½ mark)*

Step 4: Find integer n and write Molecular Formula:

$$n = \frac{MolarMass}{EFM} = \frac{188}{94} = 2$$

$$MolecularFormula = (CH_2Br)_2 = C_2H_4Br_2(1,2 - dibromoethane)$$

- *(½ mark)*

(b) Draw the structure of compound V. *(1 mark)*

```
     H   H
     |   |
H - C - C - H   (or 1,2-dibromoethane structural formula)
     |   |
    Br  Br
```

- *Correct structural formula showing all single bonds and atoms clearly.* **(1 mark)**

Question 6: Fractional Distillation of Alkanols

Separation of Propanol (b.p. 97°C), Ethanol (b.p. 78°C), and Butanol (b.p. 117°C).

(a) State the purpose of:

- **(i) Glass beads:** To increase the surface area for condensation and evaporation cycles, allowing efficient fractionating/separation of vapors of different boiling points. **(1 mark)**
- **(ii) Porcelain chips:** To ensure smooth boiling and prevent bumping/spitting of the liquid mixture. **(1 mark)**

(b) Outline the order in which the alkanols were obtained and give a reason:

- **(i) Order of separation: Ethanol → Propanol → Butanol** *(1 mark)*
- **(ii) Reason:** Liquids distill off in increasing order of their boiling points (Ethanol has the lowest boiling point of 78°C, followed by Propanol at 97°C, and Butanol highest at 117°C). *(1 mark)*

Question 7: Electrolysis of Aqueous Copper(II) Sulfate using Graphite Electrodes

Graphite electrode Q (Cathode, negative) and Graphite electrode R (Anode, positive).

(a) Write the equation for the reaction that takes place at:

(i) Electrode Q (Cathode - Negative electrode):

$$Cu^{2+}(aq) + 2e^{-} \rightarrow Cu(s)$$

- **(1 mark)**

(ii) Electrode R (Anode - Positive electrode):

$$4OH^{-}(aq) \rightarrow 2H_2O(l) + O_2(g) + 4e^{-}$$

OR

$$2H_2O(l) \rightarrow O_2(g) + 4H^{+}(aq) + 4e^{-}$$

- **(1 mark)**

(b) State one observation made in the electrolyte and give the reason. *(1 mark)*

- **Observation:** The blue color of the copper(II) sulfate solution fades / becomes colorless. **(½ mark)**
- **Reason:** Copper(II) ions (Cu^{2+}) which give the solution its blue color are continuously discharged/deposited as copper metal at electrode Q and are not replenished from the inert graphite anode. **(½ mark)**

Question 8: Group 2 Elements (Alkaline Earth Metals) & Applications

(a) Explain why group 2 elements are called alkaline earth metals. *(1 mark)*

- Their oxides react with water to form alkaline solutions (hydroxides), and their oxides/compounds are found in the earth's crust. **(1 mark)**

(b) Explain the following:

- **(i) Magnesium oxide is used in lining furnaces:** It has a very high melting point (~2852°C) and is thermally stable/refractory (does not decompose at furnace temperatures). **(1 mark)**
- **(ii) Hydrated calcium sulphate is used in making casts in hospitals:** Plaster of Paris (hydrated calcium sulfate / $CaSO_4 \cdot \frac{1}{2}H_2O$) sets hard when mixed with water and expands slightly, providing rigid support to immobilize fractured bones. **(1 mark)**

Question 9: Investigation of Products of a Burning Candle

Complete and label the diagram to show that the products are water and carbon(IV) oxide. *(3 marks)*

- **Diagram Setup Requirements & Marking Scheme:**
  - **Suction/Filter Pump setup:** Tube connected to a suction pump at the outlet to draw combustion gases through the apparatus. **(1 mark)**
  - **U-tube in Ice Bath (for Water test):** Gases pass into a cooled U-tube immersed in

ice water to condense water vapor into liquid water (tested with anhydrous cobalt(II) chloride or copper(II) sulfate). **(1 mark)**

- **Boiling Tube/Flask with Limewater (for CO_2 test):** Gases then bubble into a tube containing lime water (calcium hydroxide solution), which turns white/milky in the presence of carbon(IV) oxide. **(1 mark)**